TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Separation of Groups I, II, IIIA, IIIB, IV and V

The full separation scheme performed in order — filtrate discipline, the reason each condition is imposed, and the table that carries a mixture from one solution to six clean fractions.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 4 · LEVEL 1 TERM 1

1. Aim and working principle

Aim: to separate the basic radicals of a mixture into their analytical groups. The scheme (whose reagents were introduced in Experiment 2) works because each step removes exactly one family of cations as an insoluble compound while the conditions — acidity, common ions, buffers — are tuned so no later family can fall early. The instrument of the whole scheme is the **filtrate chain**: precipitate → filter → the filtrate (never the precipitate) receives the next reagent. A separation is only as good as its filtrations: incomplete washing or a missed precipitate poisons every later group.

2. The separation table

STEP	TO THE SOLUTION ADD	PRECIPITATE (GROUP)	CONDITIONS THAT MAKE IT WORK
0	Test the original salt for NH_4^+ (NaOH, warm) — before any reagent is added	—	NH_4^+ belongs to Group V but is tested first, as Experiment 2 explains
1	Dilute HCl, dropwise until precipitation is complete	Group I: $PbCl_2$, AgCl, Hg_2Cl_2 (white)	Filtrate → step 2. Wash the precipitate with dilute HCl water
2	Pass H_2S through the warm, dilute-HCl filtrate	Group II: sulphides — PbS black, CuS black, CdS yellow, HgS black, SnS brown, Sb_2S_3 orange…	The medium must stay acidic (about 0.3 M HCl): acid suppresses S^{2-} concentration so only the very insoluble Group-II sulphides fall — dilute or boil off excess H_2S before step 3

3	Boil off H_2S; add NH_4Cl (solid, ~1 g), then NH_4OH until alkaline	Group IIIA: $Fe(OH)_3$ brown, $Al(OH)_3$ white, $Cr(OH)_3$ green	NH_4Cl is essential: its common ion (NH_4^+) holds down OH^- so that Mg^{2+} and Group-IIIB hydroxides do not precipitate here
4	Pass H_2S through the alkaline filtrate (or add H_2S water)	Group IIIB: ZnS white, MnS flesh/pink, CoS black, NiS black	Alkaline medium: high S^{2-} concentration lets these more soluble sulphides fall
5	Boil off H_2S; add NH_4Cl + NH_4OH, then $(NH_4)_2CO_3$ solution	Group IV: $CaCO_3$, $SrCO_3$, $BaCO_3$ (white)	NH_4Cl present to keep $MgCO_3$ (Group V) in solution
6	The final filtrate holds Group V	No precipitate — test portions individually	Mg^{2+} (phosphate test), Na^+ and K^+ (flame) — Experiment 10

3. Why the conditions are what they are

- **Acid strength in step 2:** H_2S is a weak acid; in 0.3 M HCl the S^{2-} concentration is minute, and only sulphides with extremely small solubility products (Group II) can exceed them. Let the solution go nearly neutral and Group IIIB sulphides crash the party.
- **NH_4Cl before NH_4OH in step 3:** the common ion NH_4^+ suppresses NH_4OH's ionisation; the resulting low OH^- precipitates only the very insoluble IIIA hydroxides, holding $Mg(OH)_2$ and IIIB hydroxides back. Omit the NH_4Cl and the separation fails in both directions.
- **Boiling off H_2S between steps:** leftover H_2S would precipitate IIIB sulphides inside step 3's alkaline conditions and contaminate the IIIA precipitate.
- **Washing precipitates** with the wash liquid named for each step (dilute HCl water for I, H_2S water for II…) prevents both loss (peptisation) and contamination of the next filtrate.

4. Flow summary

Original solution → (+dil. HCl) → **Ppt I** | filtrate → (H_2S, acid) → **Ppt II** | filtrate → (boil; NH_4Cl + NH_4OH) → **Ppt IIIA** | filtrate → (H_2S, alkaline) → **Ppt IIIB** | filtrate → (boil; NH_4Cl + NH_4OH + $(NH_4)_2CO_3$) → **Ppt IV** | filtrate → **Group V in solution**. Memorise the chain as a sentence — "acid chloride, acid sulphide, ammoniacal hydroxide, ammoniacal sulphide, ammoniacal carbonate" — and the scheme can be rebuilt anywhere.

5. Precautions

- Test completeness of precipitation at each step (a drop of reagent in the clear filtrate should give no further cloudiness) before filtering.
- Keep every precipitate and filtrate labelled — the classic sessional disaster is two unlabelled beakers.
- Pass H_2S in the fume cupboard; it is poisonous and deadens the sense of smell, so “I can’t smell it” is not evidence of safety.
- Volumes grow at every step (washings, reagents): start concentrated enough that Group V is still detectable at the end.

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Write the filtrate chain in one line, naming each group’s precipitate.
2. Why must the Group-II precipitation be done in acid of about 0.3 M, and what error follows if the solution is made alkaline first?
3. What is the role of NH_4Cl in Group IIIA, and which unwanted precipitate appears if it is omitted?
4. Why is H_2S boiled off before Group IIIA is precipitated?
5. A student adds $(NH_4)_2CO_3$ directly to the original solution and obtains a large white precipitate. Why is this not Group IV?
6. State the wash liquids for Group I and Group II precipitates and the purpose of washing in the separation context.

Answer key

1. Solution $\rightarrow$(dil. HCl) Ppt I chlorides $\rightarrow$(H_2S/HCl) Ppt II sulphides $\rightarrow$(NH_4Cl + NH_4OH) Ppt IIIA hydroxides $\rightarrow$(H_2S/NH_4OH) Ppt IIIB sulphides $\rightarrow$($(NH_4)_2CO_3$) Ppt IV carbonates $\rightarrow$ filtrate = Group V.
2. Acid suppresses the ionisation of H_2S, holding $[S^{2-}]$ so low that only the extremely insoluble Group-II sulphides precipitate. If alkaline, $[S^{2-}]$ is high and Group-IIIB sulphides (ZnS, MnS, CoS, NiS) precipitate together with Group II — the groups merge and cannot be separated afterwards.
3. NH_4Cl’s common ion suppresses OH^- from NH_4OH so that only the least soluble hydroxides (IIIA) fall; omitted, $Mg(OH)_2$ and IIIB hydroxides (e.g. $Mn(OH)_2$, $Zn(OH)_2$) contaminate the IIIA precipitate and are lost to their own group.
4. Residual H_2S in the alkaline IIIA medium would precipitate Group-IIIB sulphides at once, contaminating the IIIA precipitate and depleting IIIB.

5. Because nothing has been removed: the carbonate would precipitate not only Ca/Sr/Ba but also most other cations as carbonates or hydroxides (the medium is uncontrolled, Mg included). Group IV exists only as the fraction remaining after Groups I–IIIB are gone and NH_4Cl holds magnesium back.
6. Group I: wash with water containing a little dilute HCl. Group II: wash with H_2S water. Purpose: remove the mother liquor (which holds the next groups' cations) without dissolving or peptising the precipitate — protecting both the precipitate's purity and the filtrate chain.
